SUPPORTING ADULT GUIDANCE

Ratio, sharing and the unitary method

Percentages, Ratio, Growth and Bounds · Lesson 1. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

Today Nutrition leads: its panel holds the lesson's main activity, for the first time in the course. In the Connection panel, the Recipe Bar shows a recipe (a smoothie, flapjack or trail mix) as a bar of equal parts. Learners change the batch size (in steps that keep every amount exact) and see one part and each ingredient. Writing the recipe as × 2 or × 5 splits the bar into more, smaller parts, while every amount and the colour of the mixture stay the same: this is why equal ratios simplify. Six questions and a stretch follow. The Nutrition panel, Read the label, uses example food labels given per 100 g: learners choose a portion, find the amount per 10 g and scale it, then reveal to check. Seven questions take this further: comparing two foods fairly, scaling a pancake recipe from 4 to 6 people, and sharing the shopping cost in a ratio. Movement has quick shares, Reflection has true-or-not statements, and Creativity is a recipe designer with six briefs.

Driving Question: *How can one part tell me everything?*

The mathematics, in plain terms

A ratio a : b : c divides a quantity into a + b + c equal parts. The unitary method finds the value of one part (or one unit) and scales. Equal ratios (multiplying every part by the same factor) describe the same proportions, so a ratio has a unique simplest form, found by dividing by the highest common factor. Per 100 g values on food labels are a unit rate; the amount in a portion of m grams is (value ÷ 100) × m.

The worked method the learner sees

These are the two worked examples the learner meets on screen, with the lesson's exact method and notation. They are here for consistency, not because the mathematics is demanding — whether you teach mathematics or are meeting it afresh alongside the learner, working from the same steps and the same language keeps your support aligned with what is on the screen, and lets you point back to a line the learner has already seen. Skip them freely if the method is already familiar.

Example 1 — Share a smoothie in the ratio 2 : 1 : 1

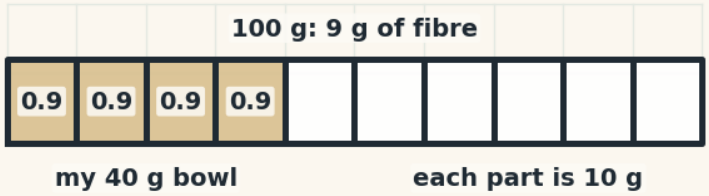
Make 600 ml of smoothie in the ratio fruit : yoghurt : milk = 2 : 1 : 1. How much of each?

1. Add the parts.	2 + 1 + 1 = 4 parts
2. One part	$\frac{600}{4} = 150$ ml
3. Fruit: 2 parts	2 × 150 = 300 ml
4. Yoghurt and milk: 1 part each	150 ml each
5. Check	300 + 150 + 150 = 600 ml

Fruit 300 ml, yoghurt 150 ml, milk 150 ml.

Example 2 — Read a label: per 100 g to my portion

Porridge oats have 9 g of fibre per 100 g. How much fibre is in a 40 g bowl?



1. 100 g is 10 lots of 10 g. per 10 g: $\frac{9}{10} = 0.9$ g
2. 40 g is 4 lots of 10 g. $4 \times 0.9 = 3.6$ g

A 40 g bowl has 3.6 g of fibre.

Recipes first, then labels

Learners meet ratio as parts of a recipe they can see before they calculate. The same move (find one, then scale) then reads food labels, scales recipes for more people and shares costs, so the lesson's mathematics is practised in the decisions it supports.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing
Connection	The Recipe Bar: one part, each ingredient, and equal ratios. Six questions. Stretch: why equal ratios make the same mixture.	~12 min
Movement	Quick shares.	~5 min
Reflection	True or not? Ratios multiply; they do not add.	~5 min
Creativity	Design a recipe card to six briefs.	~8 min
Rest	One part at a time.	~2 min
Nutrition (main)	Read the label: per 100 g to a portion, comparing fairly, scaling for more people, sharing costs.	~13 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Dividing by the wrong number of parts (for 2 : 1 : 1, dividing by 3).
Ask: “How many equal parts are in the bar altogether?”

Giving one part as the answer.
Ask: “How many parts does this ingredient have?”

Reversing the order of a ratio.
Ask: “Which ingredient comes first on the card?”

Adding the same amount to each part to “keep” a ratio (2 : 3 to 4 : 5).
Ask: “Does the second amount still compare with the first in the same way?”

Scaling a label by 100 instead of finding per 10 g or per 1 g (giving 36 g for 40 g of oats).
Ask: “Is a 40 g bowl more or less than 100 g? So more or less than 9 g?”

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line
Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson’s Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: highest common factors, a fraction of an amount, and money in pounds and pence. A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Suppose a learner says that 120 g of raisins in a 4 : 3 : 1 trail mix means there are 480 g of oats. The aim is a question that brings back the parts.

- “How many parts are the raisins?”

- “So how much is one part?”
- “How many parts are the oats?”
- “Does your answer make the oats a bit more than the raisins, as the ratio says?”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

Why is Nutrition the main Cornerstone today?

Ratio and the unitary method are what make food labels, recipes and shared bills manageable. The Nutrition panel uses them for exactly that, so the learning feeds something practical.

Food can be a sensitive subject. How is that handled?

There is no calorie counting, no weight talk, and no food is called good or bad. Labels are read only to find out what is in a portion. A dairy-free swap is offered, and ingredients are cheap and everyday. If a learner finds food difficult, the Nutrition questions can be done with the label values alone, without discussing their own eating.

Are the labels real?

They are example labels with values typical of these foods, chosen so that every portion works out exactly.

What comes next?

Lesson 2: direct and inverse proportion, which turns “find one, then scale” into an equation with a constant.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Approaching Proficiency → Developing Proficiency → Proficiency → Refining Proficiency across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like

Approaching Proficiency	Beginning to engage; names some features and makes a start with support.
Developing Proficiency	Carries out the method with prompts; can explain part of the reasoning.
Proficiency	Works through independently and explains why each step is true.
Refining Proficiency	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

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